Post-hoc ANOVA, testing assumptions for LM

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Outline

Multiple comparison error handling

Assume all assumptions for a One-way ANOVA are met, and a submodel test rejected the null hypothesis that all (subgroup means) are equal. There are several (viable) approaches to test the pairwise differences:

Fisher's Since you rejected the submodel for ANOVA, you should check for all significant differences at original α level (Least Significant Difference - LSD)

Tukey's The initial test does not convey enough evidence so joint probability of type I. error should be original α (Honest Significant Difference - HSD).

Special Use apriori known information to reduce the necessary amount of comparisons. Dunnet's (one of the groups is control group - reference for all), Hsu's (it is clear that higher/lower values are better)

LSD

If you want to use Fisher's LSD, you just need to repeatedly compute "modified t-test" for 2 independent samples at a given significance level α. So to compare groups j, l test $H_0 : \mu_j = \mu_l$, and if H_0 holds:

$$T = \frac{\bar{Y}_j - \bar{Y}_l}{s_{res}\sqrt{\left(\frac{1}{n_j} + \frac{1}{n_l}\right)}} \sim t(n-k)$$

difference to ordinary t-test for 2 independent samples (with the same variance) is in using pooled estimate for σ from all groups.

HSD

To use Tukey's method, it is necessary to define "Studentized range" and its probability distribution.

Studentized range, Tukey's HSD

Let $X_1, ..., X_k$ be independent normally distributed variables $X_i \sim N(\mu_i, \sigma^2)$, let s^2 be an independent estimator for σ^2 with ν degrees of freedom, then:

- $Q = \frac{max(X_i) - min(X_i)}{s}$ is a Studentized range with parameters k, ν and distribution $(q_{k,\nu})$
- now assume that you can get n_i observations for each X_i. Under $H_0 : \mu_1 = ... = \mu_k$ it holds for all j, l:
  $$P\left(|\bar{X}_j - \bar{X}_l| < s_{res} q_{k,n-k}(\alpha)\sqrt{\frac{1}{2}\left(\frac{1}{n_j} + \frac{1}{n_l}\right)}\right) < 1 - \alpha$$

Scheffé's comparisons

Theorem

Assume that all ANOVA assumptions hold, then if $H_0 : \mu_j = \mu_l$ holds, the following fracion follows Fisher-Snedecor distribution.

$$\frac{\frac{(\bar{Y}_j - \bar{Y}_l)^2}{k-1}}{s^2_{res}(\frac{1}{n_j} + \frac{1}{n_l})} \sim F(k-1, n-k)$$

Note: Sheffé's comparisons are generally weaker than Tukey's, but handle joint α the same way (Family Wise Error Rate is $\leq \alpha$).

Comparing different error handling methods Python example

Testing for equal variances

Bartlett's test

Assume that all groups from One-way ANOVA follow a normal distribution with the same variance σ^2. Let for each category $j \in \{1, 2, .., k\}$ be $s_j^2 = \frac{1}{n_j - 1}(\sum_{i=1}^{n_j} Y_i^2 - n_j \bar{Y_j}^2)$ and let's define $C = 1 + \frac{1}{3(k-1)}((\sum_{j=1}^{k} \frac{1}{n_j - 1}) - \frac{1}{n-k})$, then (if all $n_j > 6$)

$$B = \frac{1}{C}\left[(n-k)ln(s_{res}^2) - \sum_{j=1}^{k}(n_j - 1)ln(s_j^2)\right]$$

follows approximately $\chi^2(k-1)$ distribution.

Note: Bartlett's test is very sensitive to the normality assumption, in you need to compare variances from any continuous distribution use Levene's test (One-way ANOVA on mean absolute deviations from group means).

Testing for normality of residuals

- Comparing skewness and (excess)kurtosis to normal distribution (Jarque-Berra, Omnibus)
- Utilizing empirical distribution function (Cramér-von Mises tests "family" e.g. Kolmogorov-Smirnov, Liliefors, Anderson-Darling)
- Utilizing order statistics (Shapiro-Wilk)
- Discretizing and reformulating the problem as multinomial distribution (Pearson's goodness of fit test)

Skewness and Excess kurtosis

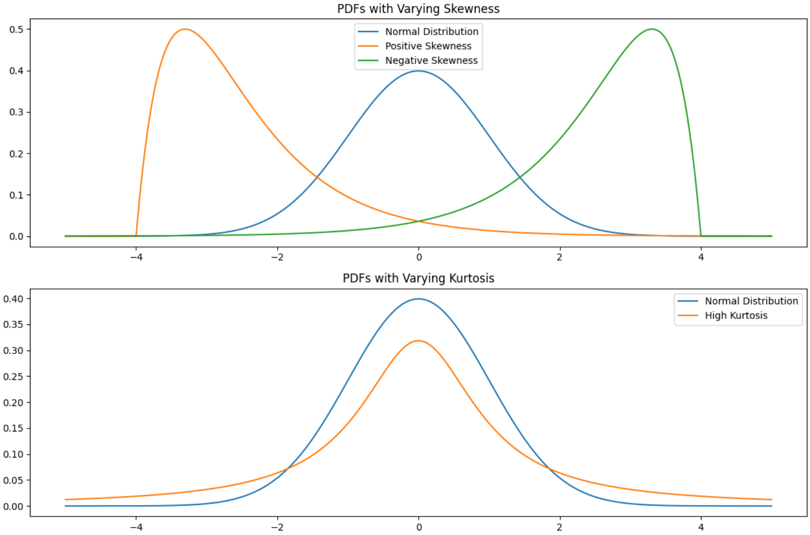
Let $x_1, ...x_n$ be a random sample from a variable with finite moments up to order 4, then skewness is defined as:

$$Skew = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^3}{(\frac{1}{n}\sum(x_i - \bar{x})^2)^{\frac{3}{2}}}$$

and excess kurtosis as:

$$Kurt = \frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^4}{(\frac{1}{n}\sum(x_i - \bar{x})^2)^{\frac{4}{2}}} - 3$$

Note: Normal distribution has both skewness and excess kurtosis equal to 0.


PDFs with Varying Skewness
Normal Distribution
Positive Skewness
Negative Skewness
0.0
0.1
0.2
0.3
0.4
0.5
−4
−2
0
2
4
PDFs with Varying Kurtosis
Normal Distribution
High Kurtosis
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
−4
−2
0
2
4

Jarque-Berra test

Assume that $X_1, \ldots X_n$ are IID normally distributed random variables. Then

$$JB = \frac{n}{6}\left(Skew^2 + \frac{1}{4}Kurt^2\right)$$

asymptotically follows $\chi^2(2)$ distribution.

Note: This time recommended n for the asymptotic distribution to hold is $n > 2000$. For lower n it is recommended to use Omnibus test. Jarque-Berra test increases type I error over specified α for "small" smaple sizes.

Two-way ANOVA

Example: Think of a way to incorporate another factor into ANOVA linear model (e.g. you can "expect" that mean wage to depend on location of your employer and "type" of work you are doing).
How will the ANOVA table change? Will the results of sequential simplifying of your model change with the order that you perform tests?

Interaction

Interaction is a simultaneous effect of two (or more) factors, that can not be explained using any of the factors by themselves.
Python example